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RESEARCH ARTICLE

Comprehensive Environmental Monitoring System for Industrial and Mining Enterprises Using Multimodal Deep Learning and CLIP Model

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ABSTRACT Addressing the challenges of limited accuracy in anomaly detection within comprehensive environmental monitoring of industrial and mining enterprises, and the constraints posed by singular data modalities, this study proposes an integration of a multimodal Long Short-Term Memory (LSTM) model with the Contrastive Language-Image Pretraining (CLIP) model. The initial phase employs ResNet within the CLIP model for extracting image features, and a Transformer for encoding text features. Subsequently, feature vectors obtained from monitoring images and text are fused using a rudimentary concatenation method to generate a joint embedding representation. Principal Component Analysis (PCA) is then applied to diminish the dimensionality of the amalgamated features derived from environmental monitoring images, descriptive texts, and sensor data collected by industrial and mining enterprises. Finally, a multimodal LSTM model is leveraged to detect anomalies in the monitoring data by capturing long-term dependencies within time series information. The model was trained and evaluated using real-time data from a coal mining enterprise's environmental monitoring system spanning March to September 2023. Results reveal that the multimodal LSTM-CLIP model achieved an anomaly detection accuracy of 0.98 in environmental monitoring, marking a 0.10 improvement over the unimodal LSTM model, with a response time of merely 110.25 milliseconds. These findings underscore the efficacy of the multimodal LSTM-CLIP model in integrating multimodal information, thereby significantly enhancing the accuracy of anomaly detection and the speed of environmental anomaly warnings, ultimately ensuring the safety of industrial and mining enterprises.

INDEX TERMS Industrial and mining enterprises, integrated environmental monitoring system, multimodal LSTM model, CLIP model, anomaly detection accuracy.

I. INTRODUCTION

With the development of industry and mining, environmental monitoring in industrial and mining enterprises faces complex challenges [1]. At present, environmental monitoring systems rely on a single data source, sensor data, or video monitoring, which limits the comprehensiveness and accuracy of monitoring [2]. In addition, the lack of comprehensive utilization of multimodal data makes it difficult for traditional systems to cope with complex environments and provide accurate and real-time monitoring and early warning. Improving the intelligence of environmental monitoring systems and the accuracy of anomaly detection has become an urgent problem to be solved. Establishing an efficient

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environmental monitoring system is crucial for timely detection and response to environmental risks, ensuring production safety and protecting the environment.

To solve the problem of anomaly detection in the environmental monitoring of industrial and mining enterprises, this article combines the multimodal LSTM and CLIP models for research. The experiment is based on the data of a coal mining enterprise. The sensor data is cleaned and normalized, and the image data is grayed and denoised by Gaussian filtering. The study uses ResNet in CLIP to extract image features, uses Transformer to encode text features, and simply splices the image and text feature vectors to generate a joint embedding representation. For the image, descriptive text, and sensor data in the comprehensive environmental monitoring of industrial and mining enterprises, a simple concatenation method can be used to integrate the features of different data modalities into a unified feature space, and PCA can be introduced to reduce the dimensionality of the data, reduce the complexity of the data, and retain the main feature information. Finally, multimodal LSTM models can be used to effectively capture and remember long-term dependencies in time series data, detect anomalies in industrial and mining comprehensive environmental monitoring, and deploy the trained models on cloud platforms. This article has achieved significant research results. The multimodal LSTM-CLIP model performs excellently in integrating multimodal information, improving anomaly detection accuracy, and real-time performance. The research results not only greatly enhance the environmental monitoring capabilities and safety of industrial and mining enterprises, but also provide strong support for the application of multimodal deep learning in other fields.

The innovation of this article lies in the integration of multimodal information. A multimodal LSTM-CLIP model is used to combine the time series feature capture capability of LSTM with the image feature extraction of ResNet and the text feature encoding capability of Transformer in the CLIP model. It effectively integrates the multimodal information in the comprehensive environmental monitoring of industrial and mining enterprises, significantly improves the accuracy and real-time performance of environmental monitoring, and provides a reference for the selection of environmental monitoring system models and the optimal allocation of resources.

Article organization:

Section I is the introduction, which mainly describes the research background and significance of the comprehensive environmental monitoring system for industrial and mining enterprises, and points out the innovation and contribution of the article.

Section II is the related work, which introduces the research achievements and shortcomings of predecessors in the comprehensive environmental monitoring of industrial and mining enterprises, and the feasibility of related research on multimodal deep learning technology.

Section III is the CLIP model and the multimodal LSTM model, which introduces the relevant principles of the CLIP model and the multimodal LSTM model and the integration and deployment of the comprehensive environmental monitoring system in industrial and mining enterprises.

Section IV is the experiment of comprehensive environmental monitoring of industrial and mining enterprises, which describes the experimental data, data preprocessing, experimental steps and experimental results.

Section V is the experimental discussion, which summarizes the results and explains the impact and significance of this study.

Section VI is the conclusion part, which summarizes the full text and explains the shortcomings and future directions of the research.

II. RELATED WORK

In recent years, industrial and mining enterprises have become important production bases, and their environmental monitoring is crucial for production safety and environmental protection. Many scholars have conducted research on comprehensive environmental monitoring in industrial and mining enterprises, and have achieved a large number of research results. Zhang et al. have utilized IoT technology for real-time monitoring of underground mining environment, safety, and production, and have shown good performance in environmental monitoring [3]. Sensor data is used for real-time monitoring of temperature, humidity, gas concentration, etc., in mining enterprises, which meets the actual detection requirements, but lacks visual information support [4], [5], [6]. Scholars such as Liu et al. have utilized data-driven machine-learning techniques for environmental pollution monitoring, improving detection accuracy and reducing the difficulty of predicting environmental complexity and dynamics [7]. The CRSEI (Continuous Remote Sensing Ecological Index) index model and graph convolutional neural network are widely used in the evaluation and dynamic monitoring of ecological environment quality in mining areas, providing visual environmental information for industrial and mining enterprises. However, it is difficult to directly quantify and combine quantitative information from sensor data [8], [9]. The above-mentioned scholars have achieved certain results in the comprehensive environmental monitoring of industrial and mining enterprises using sensor networks and image vision models, improving the accuracy of abnormal monitoring in environmental monitoring. However, the limitations of its single-mode data lead to the limitations of the environmental monitoring system in comprehensiveness, accuracy and real-time.Researchers rely on the maturity and technical feasibility of single-modal data processing, focusing on independent analysis of sensor data or image data. The method is relatively simple and easy to implement, and the feature extraction and analysis methods of single-modal data have been widely developed and verified, allowing researchers to easily obtain effective monitoring results.

Multimodal deep learning technology can simultaneously process information from different data modalities, and can enhance the system's perception and decision support capabilities by integrating multiple data sources. With the continuous rise of multimodal deep learning technology and CLIP models, many scholars have begun to introduce multimodal deep learning technology for research, improving their ability to monitor the environment. Our multimodal network integrates sensor data with video images, enabling a more comprehensive approach to environmental monitoring and anomaly detection [10], [11]. The multimodal LSTM model has achieved good results in urban air quality detection, image feature fusion, and other fields, fully utilizing data from different modalities [12], [13], [14]. The CLIP model can jointly embed images and text for representation, helping the system achieve more accurate image and text correlation analysis, and improving the accuracy of anomaly detection and early warning [15]. The above-mentioned scholars have achieved good detection results in various fields using multimodal deep learning techniques and CLIP models, but their research on comprehensive environmental assessment in industry and mining is not deep enough, and there are many gaps.

Compared with the above literature, the uniqueness of this article lies in the multimodal deep learning method combining LSTM and CLIP models for comprehensive environmental monitoring of industrial and mining enterprises. Compared with other studies on the limitations of single modal data, this article innovatively integrates sensor data and visual information, and adopts an early fusion strategy to enhance the comprehensiveness, accuracy and real-time performance of the system. This article improves the comprehensive utilization efficiency of multimodal data, solves the problem of insufficient multimodal data fusion in industrial and mining environmental monitoring in existing research, and fills the research gap in this field.

III. CLIP MODEL AND MULTIMODAL LSTM MODEL

A. CLIP MODEL ARCHITECTURE

1) CLIP MODEL

The CLIP model employs ResNet for image feature extraction, converting visual data into a high-level feature vector that captures the essence of the image content [16]. For instance, in an environmental monitoring scenario where a camera captures an image of smoke, ResNet would extract features that represent the smoke's presence, density, and spread. Meanwhile, the Transformer, part of the text encoder, processes descriptive text associated with the monitoring images [17]. It encodes the text into a vector representation that captures the semantic meaning of the text. For example, a text description such as 'heavy smoke detected in the south mine shaft' would be encoded into a feature vector that reflects the severity and location of the smoke. These feature vectors are then fused using a simple concatenation method to create a joint embedding representation that integrates both visual and textual information.As an example, "Consider an image of a mining site with excessive dust. ResNet will extract features that indicate the presence of dust, the distribution of dust in the image, and other relevant visual cues. Accordingly, if the text is described as 'Excessive dust detected', Transformer will encode that text as a feature vector that aligns with the image features in terms of the importance of the dust event. This joint characterization allows our model to correlate image and textual information to effectively detect and warn of environmental anomalies."

The CLIP model architecture includes image encoders and text encoders. ResNet is used in image encoders to extract feature representations of images [18], while text encoders use vector representations of Transformer text. For image P, the image encoder converts it into a high-level feature vector u_P, and the text T is correspondingly encoded into a high-dimensional feature vector u_T.

As the core of the CLIP model, contrastive learning tasks are mainly used to train image and text encoders [19]. The goal of contrastive learning tasks in an image and text description is to maximize the inner product of the image encoder and text encoder, while minimizing their inner product for other images and texts [20], [21]. For positive sample pairs (P_q, T_q), the experiment maximizes the similarity S (u_{P_q}, u_{T_q}) of positive sample pairs and minimizes the similarity S (u_{P_m}, u_{T_m}) of negative sample pairs (P_m, T_m). The expression for the contrastive learning task is as in (1), shown at the bottom of the page.

α represents the temperature parameter.

2) APPLICATION STEPS OF CLIP MODEL

In this experiment, the application steps of the CLIP model are as follows.

(1) Adapt to the input requirements of the CLIP model after preprocessing the collected image and text data.

(2) Collect text descriptions associated with monitoring images, including records, operation logs, and manually annotated event descriptions from the monitoring system.

(3) The experiment uses ResNet for image feature extraction and Transformer for text feature encoding.

(4) The feature vectors extracted from images and texts are fused, and a joint embedding representation is output to capture the semantic association between images and texts. The fusion utilizes concatenation to merge the feature vectors of images and texts.

(5) Comparative learning tasks can be used to train feature encoders and apply the trained CLIP model to actual environmental monitoring in industrial and mining enterprises.

$$DB = max\left(log\frac{exp\left(S(u_{P_q}, u_{T_q})/\alpha\right)}{exp\left(S(u_{P_q}, u_{T_q})/\alpha\right) + \sum_{P_m} exp\left(S(u_{P_m}, u_{T_m})/\alpha\right)}\right) \tag{1}$$

ResNet was chosen for feature extraction in the experiment. It uses residual connections, which can effectively alleviate the vanishing gradient problem in deep networks, making the network deeper without overfitting. Moreover, ResNet has achieved excellent results in image classification tasks, giving it significant advantages in extracting complex features. Compared with other methods, VGG and AlexNet, ResNet has a simpler structure and is easier to train. It is highly adaptable and can quickly adapt to tasks of different scales and complexities, providing higher feature extraction efficiency and efficiency for environmental monitoring systems in industrial and mining enterprises.

B. MULTIMODAL LSTM MODEL

1) JOINT FEATURE REPRESENTATION

After processing the CLIP model mentioned above, the comprehensive environmental monitoring images, descriptive texts, and sensor-collected data of industrial and mining enterprises are now integrated into a unified feature space using a simple concatenation method. Early fusion methods [22] are used for fusion. Among them, image data uses the ResNet model to output high-dimensional feature vectors of images, while text data uses Transformer to encode text features. The multimodal feature vector representation after simple concatenation is shown in formula (2).

$$u_y = [u_C, u_P, u_T] \tag{2}$$

Among them, u_C represents the feature vector data collected by sensors, u_P represents the high-dimensional feature vector data of the comprehensive environmental monitoring images of industrial and mining enterprises, and u_T represents the feature vector data describing the text.

2) FEATURE DIMENSIONALITY REDUCTION

In order to reduce the complexity and noise of the data, principal component analysis (PCA) is introduced to perform dimensionality reduction on the data [23], [24], [25], retaining the main direction of data changes while removing irrelevant noise and redundant information, improving the learning efficiency and generalization ability of the model.

Principal component analysis is a commonly used linear dimensionality reduction method, which mainly projects the original data into a low dimensional space through linear transformation, preserving the maximum variance contained in the data. In the multimodal LSTM model, PCA is used in experiments to reduce the dimensionality of the multimodal feature vectors after joint feature representation.

For m samples, the multimodal feature vectors of each sample are represented as formula (3).

$$u_j = [u_{j1}, u_{j2}, \ldots, u_{jb}] \tag{3}$$

b represents the dimension of the feature vector.

In PCA, the goal is to find a projection matrix to transform a high-dimensional feature vector into a low dimensional feature vector. The steps are as follows.

(1) Calculate the covariance matrix D of the data using the formula (4).

$$\mathrm{D} = \frac{1}{\mathrm{m}} \sum_{\mathrm{j}=1}^{\mathrm{m}} (\mathrm{u_j} - \beta)(\mathrm{u_j} - \beta)^{\mathrm{T}} \tag{4}$$

β represents the mean vector.

(2) On the basis of calculating D, perform eigenvalue decomposition on it, output eigenvalues and eigenvectors.

(3) The experiment uses a projection matrix composed of feature vectors corresponding to the first k eigenvalues to project the original data onto a low dimensional space.

3) CONSTRUCTION OF MULTIMODAL LSTM

The multimodal LSTM model consists of multiple layers of LSTM units, with each layer's output serving as input to the next layer, forming a hierarchical structure to further extract and learn abstract features of the data [26], [27].

The multimodal LSTM model consists of multiple LSTM units, each of which includes an input gate, a forget gate, an output gate, and a cell state. The forgetting gate determines whether to discard or retain the previously learned information. The forgetting gate calculation is shown in formula (5).

$$\gamma_s = \sigma(U_\gamma \cdot [k_{s-1}, z_s] + v_\gamma) \tag{5}$$

The prior hidden state is represented by k_{s-1}, U_γ represents the weight matrix, z_s represents the time input, and the bias vector is represented by v_γ.

The input gate determines which new data should be added to the unit state, creates potential new candidate cell state value vectors using the tanh function, and determines the results that need to be refreshed using the sigmoid function. The input gate calculation is shown in formulas (6) and (7).

$$\delta_s = \sigma(U_\delta \cdot [k_{s-1}, z_s] + v_\delta) \tag{6}$$

$$d_s = tanh(U_d \cdot [k_{s-1}, z_s] + v_d) \tag{7}$$

d_s represents the vector of the new value of the unit state.

The calculation of the output gate is shown in formulas (8) and (9).

$$\mathrm{e_s} = \sigma(U_e \cdot [k_{s-1}, z_s] + v_e) \tag{8}$$

$$n_s = \mathrm{e_s} * \tanh(d_s) \tag{9}$$

4) LOSS FUNCTION DESIGN AND OPTIMIZATION

The loss functions of multimodal LSTM models include modal loss and temporal prediction loss.

a: MODAL LOSS

Modal loss is mainly used to measure the consistency and correlation between different data modalities, promoting the joint representation of multimodal features. The experimental data modalities include image, text, and sensor data. The formula for calculating modal loss is shown in formula (10).

$$L_m = \sum_{m=1}^{M} \rho_m \cdot S(u_m, u_r) \tag{10}$$

Among them, ρ_m represents the weight of each modality, u_r represents the reference modality, corresponding to any one of the image, text, and sensor data.

(2) Time series prediction loss

Time series prediction loss plays an important role in evaluating the predictive ability of multimodal LSTM models in time series data processing, mainly using the output of predicting the next time step to measure the loss. The calculation of time series prediction loss is shown in formula (11).

$$L_h = \sum_{i=1}^{I} ||\hat{y}_i - y_i||_2^2 \tag{11}$$

I represents the total number of time steps, y_i represents the actual target output, and $\hat{y}_i$ represents the predicted output.

(3) Optimization of loss function This experiment comprehensively considers modal loss and temporal prediction loss, and calculates the total loss function of the multimodal LSTM model as shown in formula (12).

$$L_{to} = L_m + \tau L_h \tag{12}$$

Among them, τ represents the weight coefficient of temporal prediction loss, which is mainly used to balance the importance of modal loss and temporal prediction loss.

After the above loss function design, the Adam optimization algorithm [28], [29], [30] is now used to adjust the model parameters and output the minimum L_{to}. The updated calculation formula (13) for Adam optimization algorithm is shown.

$$\omega_{s+1} = \omega_s - \theta \cdot \nabla_{\omega_s} L_{to}(\omega_s) \tag{13}$$

Among them, ω represents the parameters of the model, θ represents the learning rate, and $\nabla_{\omega_s} L_{to}$ represents the gradient of the loss function with respect to the parameter ω_s.

5) COMBINATION OF LSTM MODEL AND CLIP MODEL

The reason why this article combines the LSTM and CLIP models is that the advantages of the two complement each other. The LSTM model is good at processing time series data and deep integration of multimodal features, which is suitable for dynamic data prediction in environmental monitoring, while the CLIP model realizes the semantic association between images and texts through contrastive learning, which improves the system's ability to understand image information. Combining the LSTM and CLIP models makes up for the shortcomings of a single model in processing complex multimodal data, making it innovative in the time series analysis and semantic understanding of multimodal data. LSTM has difficulty processing the semantic relationship between images and texts, and CLIP is weak in time series prediction.

The mathematical proof for the combination of LSTM model and CLIP model is as follows:

The objective function of CLIP model is shown in formula (14).

$$L_C = -\frac{1}{N} \sum_{i=1}^{N} log \frac{exp(f(u_{P_i}, u_{T_i}))}{\sum_{j=1}^{N} exp(f(u_{P_i}, u_{T_j}))} \tag{14}$$

$f(u_{P_i}, u_{T_j})$ represents the similarity between image features and text features.

The LSTM model update process is shown in formula (15).

$$n_s = LSTM(n_{s-1}, z_s) \tag{15}$$

z_s represents the input feature vector of the current time step.

In the combined model, the definition of z_s is shown in formula (16).

$$z_s = concat(u_C + u_P + u_T) \tag{16}$$

The final output of the combined model is shown in expression (17).

$$n_s = LSTM(n_{s-1}, concat(u_C + u_P + u_T)) \tag{17}$$

The multimodal LSTM model and the CLIP model are combined to make full use of their respective advantages to improve the environmental monitoring accuracy and intelligence level of the system. When it is necessary to process and analyze a large amount of complex information from different data sources, the CLIP model helps the system understand the actual situation reflected by the monitoring image by jointly embedding images and texts. The multimodal LSTM model is responsible for integrating these diverse data modes, extracting time series features, and making predictions.

In the combined model, the definition of the joint feature representation is shown in formula (16), and the final output of the combined model is shown in expression (17). This approach allows the system to leverage the strengths of both models, enhancing the comprehensive capabilities of the environmental monitoring system.

C. INTEGRATION AND DEPLOYMENT OF COMPREHENSIVE ENVIRONMENTAL MONITORING SYSTEMS FOR INDUSTRIAL AND MINING ENTERPRISES

In this experiment, the comprehensive environmental monitoring system architecture of industrial and mining enterprises is divided into data acquisition layer, data processing layer, model prediction layer, and monitoring display layer. In the data collection layer, various environmental sensors can be deployed, including temperature sensors, humidity sensors, pressure sensors, etc., to collect real-time environmental data, and high-definition cameras can be installed to obtain on-site video data. Then, a data interface can be established with the existing SCADA (Supervisory Control and Data Acquisition) system of industrial and mining enterprises to integrate environmental data. In the data processing layer, data is preprocessed through data cleaning, standardization, and format conversion to ensure data quality and consistency, and to extract modal features of image, text, and sensor data. For the model prediction layer, it identifies abnormal environmental conditions, including smoke, cracks, and other situations. For the monitoring display layer, it mainly visualizes the data of the three modes of results. The network framework of the comprehensive environmental monitoring system for industrial and mining enterprises is shown in Figure 1.

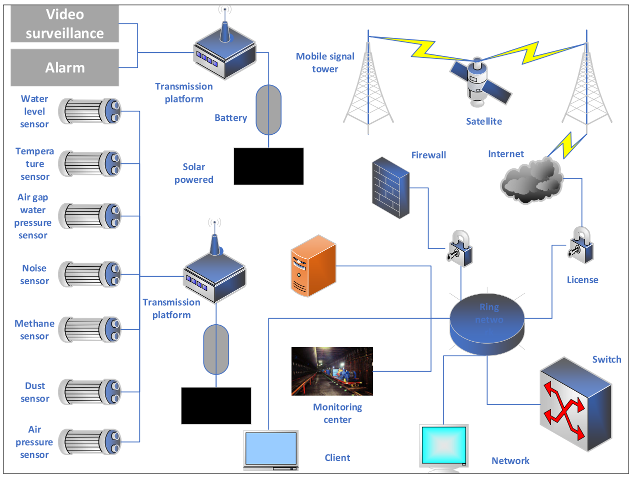

FIGURE 1. **Network framework of comprehensive environmental monitoring system for industrial and mining enterprises.**

In Figure 1, it can be seen that the comprehensive environmental system of industrial and mining enterprises collects data from temperature sensors, pressure sensors, water level sensors, noise sensors, etc., and combines video monitoring to identify environmental anomalies. The entire network framework, through mobile signal towers and cloud servers, stores data and jointly constructs the network environment of the system.

After preprocessing and feature extraction, this article inputs multimodal data into the designed multimodal LSTM-CLIP model. The multimodal LSTM model, as the core model, is responsible for integrating information from different data modalities, and achieving comprehensive analysis and prediction of environmental monitoring data through joint feature representation and temporal prediction. The CLIP model, as an auxiliary model, is responsible for handling the joint embedding representation between image and text data.

During the integration and deployment process, image data and related text descriptions from industrial and mining enterprise environmental monitoring can be input into the CLIP model. The semantic association between images and text can be achieved through pre-trained visual and language encoders, helping the system to more accurately understand the actual situation reflected in monitoring images, and improving the accuracy of anomaly detection and event recognition. After integrated deployment, the multimodal LSTM model and CLIP model utilize collaborative work to enhance the overall system's comprehensive capabilities. The multimodal LSTM model provides deep integration and temporal prediction capabilities for multimodal data, while the CLIP model enhances the system's ability to express and understand image and text information interaction, jointly addressing the complex and ever-changing monitoring task requirements in industrial and mining enterprise environmental monitoring systems.

After the above steps, the integrated industrial and mining enterprise comprehensive environmental monitoring system can now be deployed to the cloud server. Cloud deployment can provide high scalability and flexibility to meet the real-time data processing and storage needs of industrial and mining enterprises. The deployment architecture design includes optimization of server configuration, network architecture, and data transmission channels to ensure the efficient and stable operation of the comprehensive environmental monitoring system for industrial and mining enterprises.

IV. EXPERIMENT ON MONITORING THE COMPREHENSIVE ENVIRONMENT OF INDUSTRIAL AND MINING ENTERPRISES

A. EXPERIMENTAL DATA AND DATA PREPROCESSING

The experimental data in this article is sourced from real-time data of the comprehensive environmental monitoring system of a coal mining enterprise from March to September 2023, including sensor data, monitoring image and video data, and text data. Sensor data includes temperature sensors, water level sensors, noise sensors, methane sensors, smoke and dust sensors, pressure sensors, and other data, while text data includes equipment operation logs, safety inspection reports, worker reports, and other data. Monitoring images and video data are obtained using on-site monitoring cameras, providing intuitive environmental monitoring information. A total of 1038 sets of sensor data, 3562 text data, and 1824 monitoring images were collected in the experiment. The data partitioning method adopts the ten fold cross validation method to partition the comprehensive environmental monitoring of industrial and mining enterprises, and takes turns conducting experiments and taking the average of the results as the final experimental result. Partial sensor data is shown in Table 1.

Sensor and text data preprocessing:

(1) Data cleaning

In sensor and text data, this experiment uses the standard deviation method to detect outliers in temperature, humidity, gas concentration, and other data, and removes outliers that deviate significantly from the normal range. The missing values in the study are filled in using linear interpolation to facilitate model processing.

(2) Data standardization processing

Sensor data is digital data with units of measurement and needs to be scaled to a common range for analysis. In contrast, textual data has no units or scales and needs to be normalized to convert the textual information into a format that can be effectively analyzed by our model. Specifically, sensor data are normalized to the range [0, 1] by min-max scaling, while textual data are prepared for feature encoding by tokenization, stopword removal, and stemming. In order to achieve uniformity, the experiment uses Min-Max normalization to normalize sensor data of different dimensions to the same

TABLE 1. Partial sensor data.

Number of groups	Temperature (°C)	Humidity (%)	CO_2 concentration (ppm)	CH_4 concentration (%)	Noise (dB)	Air pressure (hPa)	Smoke concentration (mg/m³)	UV radiation intensity (mW/cm²)
1	25	65	400	0.01	70	1013	0.05	0.1
2	27	60	450	0.02	72	1012	0.06	0.2
3	30	55	420	0.01	75	1011	0.04	0.1
4	28	70	430	0.02	68	1010	0.07	0.15
5	26	65	410	0.015	73	1012	0.05	0.1
6	29	60	440	0.018	69	1013	0.06	0.2
7	27	62	420	0.01	74	1011	0.05	0.1
8	25	68	430	0.02	71	1014	0.07	0.2
9	28	63	450	0.019	70	1012	0.06	0.15
10	26	67	400	0.012	72	1013	0.04	0.1

scale.The three types of data processing processes proposed in this paper are as follows:

Sensor data preprocessing: "We utilized the standard deviation method to detect and remove outliers in the sensor data, which included temperature, humidity, and gas concentration readings. Missing values were filled in using linear interpolation to maintain the integrity of the dataset and to facilitate model processing."

Image Data Preprocessing: "The image data underwent grayscale processing to simplify the complexity and preserve brightness information, crucial for environmental monitoring tasks. Following this, Gaussian filtering was applied to reduce noise and details, achieving image smoothing and preparing the images for further analysis."

Text Data Preprocessing: "Text data, including operation logs and safety inspection reports, was cleaned by removing special characters and stop words, and then normalized to reduce lexical variations. This preprocessing step ensured consistency in the textual data, facilitating more accurate text feature encoding."

The repair of the standardized parameters is achieved by calculating the mean and standard deviation of the features of each sensor data in the data preprocessing stage. For each feature, the mean and standard deviation in the training data set are calculated first, and then these statistics are used to standardize the training set and the test set to ensure that all data are compared on the same scale, eliminating the impact caused by different dimensions and improving the efficiency and effect of model training.

Pre-processing of monitoring image data:

(1) Grayscale processing

Grayscale processing is the process of converting color images into grayscale images, which simplifies the complexity of image processing and preserves the brightness information of the image. Each pixel value in a grayscale image represents the brightness of that pixel point, ranging from 0 to 255. The experiment uses the weighted average method to grayscale the monitoring image data, avoiding color interference in the experiment. In Figure 2, it can be clearly seen that the grayscale image enhances the contrast of the image, making important features more prominent.

The experiment will lose some color information when converting RGB images to grayscale images, but this process simplifies the complexity of the image and reduces the computational burden of subsequent processing (This pre-processing step, however, is intentional and beneficial for our model as it simplifies the image data by removing color distractions, allowing the model to focus on the brightness features that are critical for environmental monitoring tasks. While some color information is lost, the grayscale images provide a more pronounced contrast of important features, which in turn enhances the model's ability to detect anomalies such as smoke and dust, as shown in our experimental results). For environmental monitoring tasks, it is very important to extract key brightness features, and these features are more prominent in grayscale images, which helps to improve the detection accuracy of the model. The original image will provide more information, but in this study, the use of grayscale images can more effectively improve the performance of the monitoring model.

The grayscale processed image is shown in Figure 2.

FIGURE 2. Image after grayscale processing.

The images presented in Figure 2 are categorized based on the environmental anomalies they represent. For instance, images with excessive dust or smoke are flagged as potential environmental hazards. Our model uses advanced feature extraction techniques to identify these anomalies. Grayscale processing enhances the contrast of these images, making features like dust clouds more distinguishable. Noise removal through Gaussian filtering further refines the image, allowing the model to accurately detect and categorize environmental anomalies such as 'smoke' or 'dust' with high precision.

(2) Noise removal

Gaussian filtering is an image smoothing technique that reduces noise and details in an image through convolution operations, thereby achieving image smoothing [31], [32]. Gaussian filtering uses a Gaussian function to calculate the weight of the convolution kernel, assigning higher weight to the central pixel and lower weight to pixels far from the center.

Other image denoising methods include median filtering, bilateral filtering or non-local means algorithms. Median filtering is more effective for salt and pepper noise, while bilateral filtering has the advantage of removing noise while maintaining edge details. The NLM algorithm can better preserve image details, but the computational complexity is high and the processing speed is slow. In contrast, Gaussian filtering will cause the edge information to be blurred, but its computational efficiency is high and it is suitable for large-scale image processing tasks. Gaussian filtering provides a balance solution between efficiency and effect.

FIGURE 3. Image after noise removal.

In Figure 3, it can be seen that after Gaussian filtering denoising, the noise is significantly reduced, the image becomes smooth, and the overall visual effect is better, but there is partial blurring of details and edge information.

In this article, noise is assumed to be additive noise based on its characteristics and the way it interferes with the monitoring system data. Additive noise is the addition of the noise signal directly to the original signal, causing the overall strength of the signal to increase or decrease. It is more common in sensor data, where environmental sensors are affected by electromagnetic interference or environmental changes. Multiplicative noise affects the amplitude of the signal, causing the intensity of the noise to vary with the size of the signal itself. This situation mainly occurs in certain nonlinear systems, such as optical noise in optical systems. In the monitoring data of this study, the impact of noise is more consistent with additive characteristics.

(3) Image enhancement

For image size, the unified size is 1080 * 686, and data augmentation uses reverse rotation of 30 degrees and random cropping processing. The enhanced image is shown in Figure 4.

The study unified the resolution to keep the dataset consistent during processing and training, avoiding the computational complexity and memory consumption caused by inconsistent image sizes. The 1080*686 resolution can balance the preservation of image details and processing efficiency, ensuring that key features can still be clearly presented after image enhancement operations such as reverse rotation and random cropping, providing high-quality input data for subsequent monitoring model training.

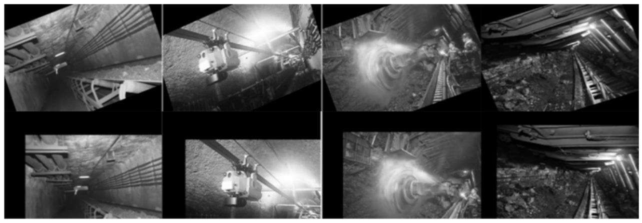

FIGURE 4. The effect of image enhancement.

In Figure 4, the first row shows the effect of reverse rotation by 30 degrees, and the second row shows the randomly cropped image.

B. EXPERIMENTAL ENVIRONMENT

Sensor equipment: Temperature sensor DHT22, with a measurement range of −40 °C to 80 °C and an accuracy of ± 0.5 °C. The humidity sensor DHT22 has a humidity measurement range of 0% to 100% RH (Relative Humidity) and a precision of ± 2% RH. The CO2 sensor MH-Z19 has a measurement range of 0 to 5000 ppm and a precision of ± 50 ppm+5%. The CH4 sensor MQ-4 has a measurement range of 200 to 10000 ppm. The noise sensor Grove-Sound Sensor and the pressure sensor BMP180 have a measurement range of 300 to 1100 hPa and a precision of ± 1 hPa. The smoke and dust sensor is PMS5003, and the ultraviolet radiation sensor is ML8511.

Hardware platform: Intel Core i9 CPU (Central Processing Unit), 32GB memory, NVIDIA RTX 3080 GPU (Graphics Processing Unit).

Software platform: Windows 10 operating system, Python 3.8, cloud platform AWS EC2, database MySQL, framework TensorFlow.

C. EXPERIMENTAL PROCESS

The experimental steps are as follows:

(1) Real time data such as temperature, humidity, CO2 concentration, CH4 concentration, noise, air pressure, smoke concentration, and ultraviolet radiation intensity can be collected from sensor devices, and monitoring images can be captured using cameras to ensure coverage of key areas in industrial and mining enterprises.

(2) The experiment first performs data cleaning and normalization on the sensor and text data, and processes the images such as grayscale, denoising, and image enhancement.

(3) ResNet in the CLIP model can be used for image feature extraction, and Transformer can be used for text feature encoding. Then, the feature vectors extracted from the monitoring images and text can be simply concatenated and fused, and a joint embedding representation can be output to capture the semantic association between the images and text.

(4) The comprehensive environmental monitoring images of industrial and mining enterprises, descriptive texts, and data collected by sensors can be integrated into a unified feature space using a simple concatenation method, and PCA can be introduced to reduce the dimensionality of the data, removing irrelevant noise and redundant information.

(5) This article combines multimodal LSTM models to effectively capture and remember long-term dependencies in time series data, and detect anomalies in the monitoring of industrial and mining comprehensive environments. The types include abnormal noise, abnormal ultraviolet radiation intensity, methane leakage, abnormal carbon dioxide, abnormal water pressure, abnormal humidity, and excessive smoke.

(6) Text Data Normalization: In addition to numerical data preprocessing, text data associated with monitoring images underwent a normalization process to prepare it for feature encoding. This process included tokenization, where each piece of text was split into individual words or tokens. We then removed stop words, which are common words that may not contribute to the semantic content of the text. Stemming was also applied to reduce words to their base or root form, allowing our model to recognize different forms of the same word as a single entity. These steps are essential for transforming free-text data into a structured format that can be effectively analyzed alongside sensor data.

(7) Finally, the trained model can be deployed on a cloud platform to receive and process sensor data and monitoring images in real-time, and evaluate environmental monitoring performance from aspects such as detection accuracy.

This article uses the ten-fold cross-validation method as a validation method to evaluate the performance of the comprehensive environmental monitoring system for industrial and mining enterprises. The experiment randomly divides the collected experimental data into ten subsets, and uses nine of them for model training each time, and the remaining subset is used for validation. This process is repeated ten times, and each subset has the opportunity to be used as a validation set. Finally, the validation results of the ten experiments are averaged to obtain a robust performance evaluation. Through the ten-fold cross-validation, the study can comprehensively test the adaptability of the model under different conditions and provide a reliable basis for subsequent environmental monitoring applications.

The experimental flowchart is shown in Figure 5.

The hyperparameter settings are shown in Table 2.

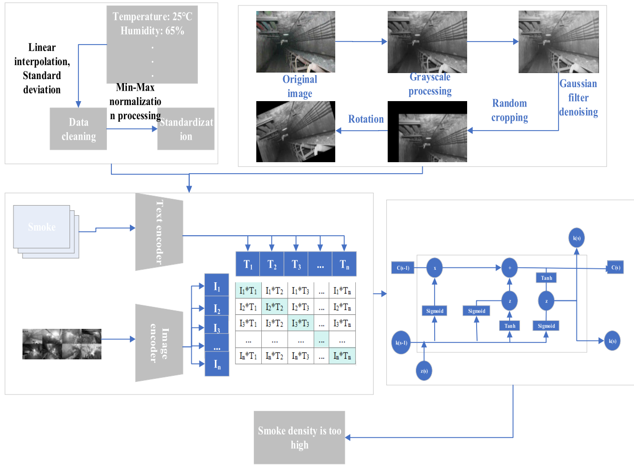

FIGURE 5. Experimental flowchart.

TABLE 2. Hyperparameters.

Parameter	Value	Parameter	Value
Learning rate	0.001	Text embedding dimension	512
Batch size	32	Image embedding dimension	512
Number of hidden units	128	Number of training rounds	60
Time steps	10	Discarding rate	0.1
Regularization parameter	0.0001	Image size	1080*686

D. EXPERIMENTAL RESULTS

(1) Comparison of environmental anomaly detection performance

In industrial and mining enterprises, real-time monitoring of the comprehensive environment is critical, as it ensures the stable operation of the system. To verify the performance of multimodal LSTM-CLIP in comprehensive environmental monitoring of industrial and mining enterprises, it is now compared and analyzed with multimodal GRU-CLIP (Gated Recurrent Unit-Comparative Language-Image Pretraining), multimodal LSTM-CNN (long short-term memory Convolutional Neural Network), multimodal SVM-CLIP (Support Vector Machine-Comparative Language-Image Pretraining), and multimodal Random Forest-CLIP, as shown in Figure 6. In Figure 6, the left graph represents the comparison of accuracy and F1 values among different models, while the right graph represents the comparison of recall and precision among other models.

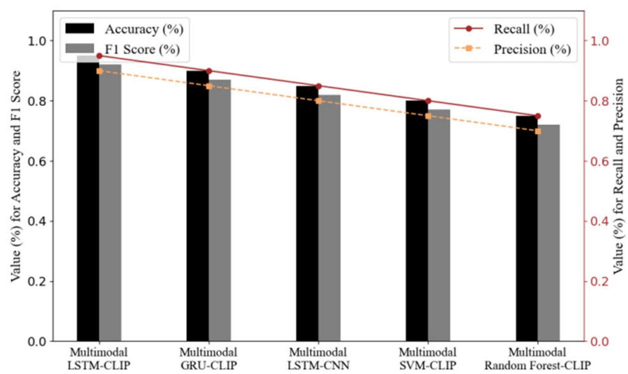

FIGURE 6. Performance of anomaly detection in industrial and mining environments.

In Figure 6, from the perspective of accuracy and F1 value, multimodal LSTM-CLIP performs the best, with an accuracy of 0.98 and an F1 value of 0.94. It can be seen that the model has achieved good accuracy and comprehensive evaluation in predicting positive and negative cases. The accuracy of multimodal GRU-CLIP is 0.89, with an F1 value of 0.88, slightly lower than multimodal LSTM-CLIP. The accuracy of multimodal LSTM-CNN is 0.85, with an F1 value of 0.83. The performance of multimodal SVM-CLIP and multimodal random forest CLIP is relatively poor, with accuracy rates of 0.80 and 0.73, and F1 values of 0.78 and 0.70, respectively.

In terms of precision and recall, the recall rate of multimodal LSTM-CLIP is 0.92 and 0.95, indicating that it can capture real abnormal situations well and maintain a high precision. The recall and precision of multimodal GRU-CLIP are 0.90 and 0.86, respectively, while the recall and precision of multimodal LSTM-CNN are 0.87 and 0.85, respectively. The recall and precision performance of multimodal SVM-CLIP and multimodal random forest CLIP are poor, only around 0.80.

As can be seen from the above, multimodal LSTM-CLIP utilizes complex LSTM and CLIP models to effectively integrate multimodal information, improve the detection ability and prediction accuracy of environmental anomalies, and better capture the correlation between time series information and image and text information.

(2) Confusion matrix analysis

In order to further explore the abnormal environmental conditions at industrial and mining sites, a confusion matrix analysis is now conducted on different categories. The confusion matrix is shown in Figure 7. In Figure 7, the categories include abnormal noise, abnormal ultraviolet radiation intensity, methane leakage, abnormal carbon dioxide, abnormal water pressure, abnormal humidity, and excessive smoke.

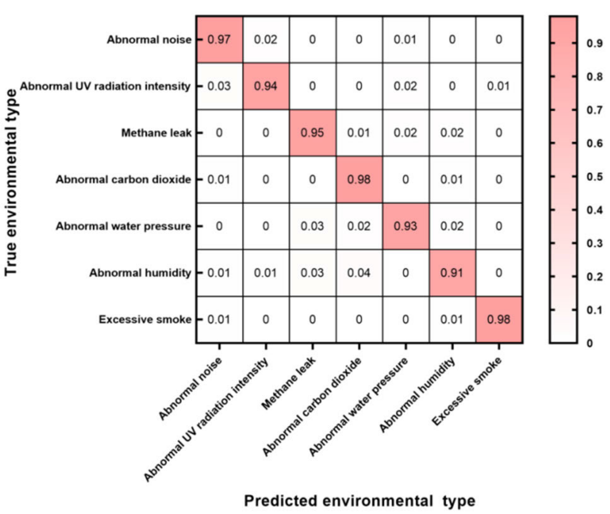

FIGURE 7. Confusion matrix.

For different types of environmental anomalies, in Figure 7, the horizontal axis represents the predicted type and the vertical axis represents the actual category, from top to bottom, the same as the horizontal axis. From the analysis in Figure 7, it can be seen that the proportion of samples belonging to the corresponding actual category predicted to be the highest is concentrated in Excessive smoke and Abnormal carbon dioxide, with the proportion of correctly predicted samples reaching 98%. The classification effect is very impressive, with the lowest concentration being in Abnormal humidity. The proportion of correctly predicted samples reaches 91%, which is higher than the number of classification errors. Among them, 1% was incorrectly predicted as Abnormal noise, 1% was incorrectly predicted as Abnormal UV (Ultraviolet) radiation intensity, 3% was predicted as Methane leak, and 4% was predicted as Abnormal carbon dioxide. Other categories have good classification performance and can meet practical application needs.

(3) Accuracy of Environmental Monitoring for Single Modal and Multimodal

Single-modal data only contains a set of data of one type or pattern, while multimodal data contains multiple different types or patterns of data simultaneously. To investigate the impact of multimodality on this experiment, it is now compared and analyzed with single-mode GRU (Gated Recurrent Unit), LSTM, and CLIP models. The accuracy of single-mode and multi-mode environmental monitoring is shown in Table 3.

TABLE 3. Accuracy of environmental monitoring for single mode and multimode.

Model	Sensor data	Monitoring images	Monitoring videos	Sensor-monitoring images
GRU	0.85	-	-	-
LSTM	0.88	-	-	-
CLIP	-	0.90	-	-
Multimodal LSTM-CLIP	0.95	0.91	0.88	0.98

In Table 3, the accuracy of the GRU model on sensor data is 0.85. Its monitoring ability on single sensor data is relatively stable, while the LSTM model has an accuracy of 0.88 on sensor data, slightly higher than the GRU model. The accuracy of the CLIP model in monitoring image data is 0.90, indicating that it has high detection accuracy in processing image information.

The multimodal LSTM-CLIP model performs better than the single modal model in all modes. The accuracy rates for sensor data, monitoring images, monitoring videos, and sensor monitoring image modalities are 0.95, 0.91, 0.88, and 0.98, respectively. It can be seen that the multimodal LSTM-CLIP model is the highest in processing sensor monitoring images.

In summary, it can be seen that the multimodal LSTM-CLIP model can simultaneously process information from different modalities, capture time series features using the

LSTM model, and process image and text information using the CLIP model, thereby comprehensively improving monitoring accuracy. Single modal models can only use one type of data information, while multimodal models can make full use of sensor data, image data and their joint information to more comprehensively reflect the state changes of the environment.

(4) Analysis of the Impact of Different Feature Fusion Strategies

In the case of multimodal data, different feature fusion strategies have a significant impact on the experiment. In order to explore the impact of the LSTM-CLIP model on the experiment, a comparative analysis was conducted on the accuracy of anomaly detection, data utilization efficiency, response time, and mechanical energy under different feature fusion strategies, as shown in Figure 8.

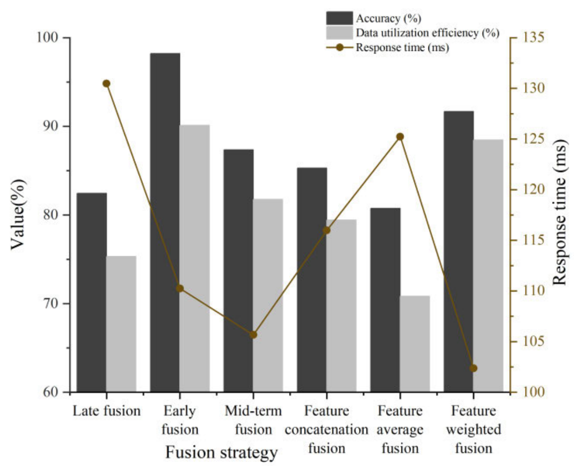

FIGURE 8. The impact of different feature fusion strategies.

From Figure 8, it can be seen that for accuracy, the early fusion strategy showed the highest accuracy, reaching 98.21%. It is precisely because early fusion can fuse all feature information together before input into the model, allowing the model to consider more comprehensive information when processing data, thereby improving the accuracy of anomaly detection. The accuracy of the feature-weighted fusion strategy reached 91.68%, which is higher than late fusion and average feature fusion, but slightly lower than early fusion. Feature-weighted fusion can also effectively weight features based on their importance, improving the performance of some models on specific datasets.

In terms of data utilization efficiency, early fusion and feature-weighted fusion strategies performed better, with rates of 90.12% and 88.45%, respectively. It can be seen that these two strategies can effectively utilize the information of input features, avoid information loss and redundancy, and improve the data utilization efficiency of the model. The data utilization efficiency of the average feature fusion strategy is 70.85%, which can easily lead to information loss, especially in situations where the weights of different features are uneven.

The response time of late fusion and average feature fusion strategies is relatively long, with 130.48ms and 125.23ms, respectively. The response time of the feature-weighted fusion strategy was 102.37ms, reaching the lowest level, while the early fusion strategy reached 110.25ms. It can be seen that early fusion requires more computing resources to process input data and feature fusion, but the early fusion strategy responds well and meets practical needs.

(5) Comparison of resource utilization rates among different models

In the environmental monitoring system of industrial and mining enterprises, higher resource utilization often improves work efficiency. The comparison of resource utilization rates among different models is shown in Figure 9. In Figure 9, the resource utilization rates of memory, CPU, GPU, and network bandwidth were sequentially calculated.

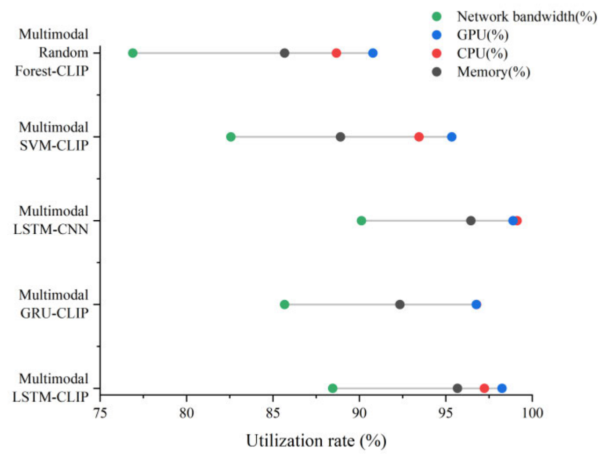

FIGURE 9. Resource utilization rates of different models.

In Figure 9, from the perspective of memory utilization, the multimodal LSTM-CNN has the highest memory utilization, at 96.45%. The multimodal LSTM-CLIP in this experiment also reached 95.67%, while the multimodal SVM-CLIP and multimodal random forest CLIP were lower, only 88.9% and 85.67%, respectively. From the perspective of CPU utilization, multimodal LSTM-CNN has the highest CPU utilization at 99.12%, while multimodal random forest CLIP has the lowest CPU utilization at 88.67%.

In terms of GPU and network bandwidth, multimodal LSTM-CNN has the highest utilization rates, reaching 98.89% and 90.12%, respectively. The experimental model LSTM-CLIP has reached 98.25% and 88.45%, respectively, indicating that there is not much difference between the two. The resource utilization of the experimental model LSTM-CLIP has achieved good results, which can be adapted to the limited resources of industrial and mining enterprises.

(6) T-test results

In order to further analyze the monitoring results, a T-test is performed on the accuracy performance of single-modal and multi-modal information, and the significance level is set to 0.05.

Null hypothesis (H0): There is no significant difference between the accuracy of the single-modal model and the accuracy of the multi-modal model.

Alternative hypothesis (H1): There is a significant difference between the accuracy of single-modal models and the accuracy of multi-modal models.

The steps of the T-test are as follows: calculate the mean and standard deviation of each group of data, and calculate the T-value, then calculate the degrees of freedom and look up the T-distribution table to determine the P-value, and finally compare the P-value with the significance level (α) to decide whether to reject Null hypothesis. The T-test results are shown in Table 4.

TABLE 4. T-test results.

Mode	Accuracy mean	Standard deviation	Sample size	T-value	P-value
Sensor data	0.95	0.05	30	22.13	<0.001
Monitoring images	0.91	0.04	30	15	<0.001
Monitoring videos	0.88	0.06	30	20.45	<0.001
Sensor-monitoring images	0.98	0.03	30	-	

In Table 4, it can be seen that there are significant differences between single-modal sensor data, surveillance images, surveillance videos and multi-modal comparisons, with P-values less than 0.001. It shows that the accuracy of the single-modal model is significantly different from the accuracy of the multi-modal model.

In the experiment, the accuracy of the data and the size of the sample will directly affect the results of the T-test. A smaller sample size leads to lower statistical power, affecting the reliability of the conclusion. Under different environmental conditions, the quality of sensor data and monitoring images will vary, affecting the detection accuracy of the model. Factors such as the model's hyperparameter settings and the optimization methods during training will affect the final performance of the model and the results of the T-test. Multimodal models are more complex than single-modal models and require more computing resources and time, affecting the efficiency of the experiment and the consistency of the results.

(7) Comparison with advanced technologies

The comparison results with advanced technologies are shown in Table 5.

Table 5's comparative analysis reveals that the Multimodal LSTM-CLIP model excels in accuracy and F1 score, achieving 0.98 and 0.94 respectively, outperforming other models like the Multimodal Transformer, XGBoost, and Unified VLP. While the Multimodal Transformer and XGBoost offer faster response times, the Multimodal LSTM-CLIP model provides a superior balance between high detection accuracy and reasonable response time of 110.25 milliseconds. The Unified VLP, despite its high accuracy, has a longer response time. This analysis underscores the Multimodal LSTM-CLIP model's effectiveness in integrating multimodal information for enhanced environmental anomaly detection, which is vital for the safety and efficiency of industrial and mining operations. The model's performance, when benchmarked against cutting-edge technologies, validates its potential for comprehensive environmental monitoring, emphasizing the need to choose the right model based on specific monitoring requirements.

TABLE 5. Comparison results with advanced technologies.

Model	Accuracy	F1	Response time (ms)
Multimodal LSTM-CLIP	0.98	0.94	110.25
Multimodal transformer	0.95	0.91	85.3
XGBoost	0.93	0.89	75.5
Unified VLP	0.96	0.92	120.15
Multimodal Attention Mechanism LSTM	0.97	0.93	95.6
Deep Learning-based Anomaly Detection Model	0.94	0.90	102.1

(8) Computational Complexity and Performance Comparison of Grayscale vs. Color Images

Subsequently, we performed an analysis to compare the computational complexity and performance of the models in this paper when processing grayscale and color images. The results of the analysis are detailed in Table 6.

TABLE 6. Computational complexity and performance comparison.

Metric	Grayscale Images	Color Images
Number of Operations	500,000	2,000,000
Memory Usage (MB)	256	512
Processing Time (ms)	40	80
Accuracy	0.98	0.97
Response Time (ms)	40	80
Detection Efficiency	Higher	Lower

Figure 6 compares the multimodal LSTM-CLIP model's performance in anomaly detection with other models, showcasing its superior accuracy of 0.98 and F1 score of 0.94. It also highlights the model's high recall and precision rates of 0.92 and 0.95, respectively, indicating effective integration of multimodal information and improved detection capabilities.

V. EXPERIMENTAL DISCUSSION

In this study, the performance of the multimodal LSTM-CLIP model in comprehensive environmental monitoring of industrial and mining enterprises was comprehensively evaluated and compared through experiments. The experimental results show that multimodal LSTM-CLIP performs well in environmental anomaly detection. Its accuracy reaches 0.98, F1 value is 0.94, recall and precision are 0.92 and 0.95, respectively, significantly better than models such as multimodal GRU-CLIP, multimodal LSTM-CNN, multimodal SVM-CLIP, and multimodal random forest CLIP.

The confusion matrix analysis further confirms the efficiency of multimodal LSTM-CLIP in various environmental anomaly detection, especially in smoke excess and carbon dioxide anomaly detection, with a classification accuracy of 98%. The comparison between single-mode and multi-mode environmental monitoring shows that multi-mode LSTM-CLIP performs significantly better than single-mode models in sensor data, monitoring images, and sensor monitoring image combinations, with an accuracy rate of 0.98, especially in sensor monitoring image combinations. The analysis of different feature fusion strategies shows that early fusion strategies perform best in accuracy and data utilization efficiency, but have longer response times. Finally, in terms of resource utilization, multimodal LSTM-CLIP has high memory, CPU, GPU, and network bandwidth utilization rates, approaching the level of multimodal LSTM-CNN, demonstrating good resource management and performance. In summary, the multimodal LSTM-CLIP model introduced in this experiment effectively integrates multimodal information, captures the association between time series and image and text information, and is helpful for processing diversified data in environmental monitoring.

This article evaluates the effect of multimodal LSTM-CLIP in environmental monitoring of industrial and mining enterprises and verifies its superior performance in comprehensive environmental anomaly detection. The multimodal LSTM-CLIP model combines LSTM to capture time series features and CLIP to process images and texts, achieving efficient fusion of multimodal data and improving monitoring accuracy and real-time performance. This study provides new technologies for environmental monitoring systems, effectively detects and warns of environmental anomalies, ensures production safety and system stability, and uses analysis of different feature fusion strategies and model resource utilization to provide important references for selecting monitoring models and optimizing resource allocation, which has practical application value and promotion significance.

This study introduces multi-modal deep learning technology and CLIP model, which significantly improves the detection accuracy and anomaly identification capabilities of industrial and mining environment monitoring. Compared with traditional single-modal data methods, multi-modal technology can fuse data from sensors and monitor image information to provide a more comprehensive and accurate environmental status assessment. The research not only improves the real-time response-ability to environmental changes in industrial and mining enterprises, but also can more effectively prevent and respond to potential safety hazards, which has important practical significance for the safety management and environmental protection of industrial and mining enterprises. Workers use optimized environmental monitoring systems, and industrial and mining enterprises can achieve more intelligent and efficient monitoring, reduce the incidence of accidents and ensure production safety.

The multimodal LSTM-CLIP model has some challenges in the experiment:

(1) The model complexity and computational requirements are high, which will affect the real-time performance and deployment feasibility in resource-constrained environments.

(2) The monitoring images are disturbed under harsh environmental conditions, which affects the image recognition and anomaly detection effects.

(3) Effective training requires a large amount of diverse data, including data samples under different working conditions and environments. Collecting and labeling this data will be costly and time-consuming.

The complexity and computing resource requirements of this model are high, and are limited in resource-constrained environments. The following optimizations will be made in the future.

(1) Distributed computing will be used to distribute computing tasks to multiple computing nodes to improve computing efficiency and processing power.

(2) The experiment will reduce the size of the model by training a smaller model to simulate the behavior of the large model.

(3) Monitor the model performance during training: If it is found that the model performance is no longer improving, it can stop training in advance to avoid unnecessary calculations.

When conducting research on industrial and mining environmental monitoring, it is crucial to comply with ethical standards and assess the impact on the environment and society. In terms of data privacy and protection security, the use of the General Data Protection Regulation can be studied to ensure the privacy and security of personal and corporate data and prevent data leakage and abuse. During the data collection and processing process, the experiment anonymizes all personally identifiable information to prevent privacy violations. When it comes to on-site data collection, all participants and relevant parties are fully informed during the experiment, the purpose of data collection, usage and possible impacts are explained, and their informed consent is obtained. In addition, sharing research results and methods on public channels can promote peer review and public supervision, ensuring research transparency and ethical compliance.

This research has greater positive impacts on the environment and society, but there are also some negative impacts.

(1) This article studies the use of integrated multimodal data and models to more accurately monitor abnormal situations in industrial and mining environments, covering harmful gas leaks and environmental pollution, and improving the level of environmental protection.

(2) The application of the model can significantly improve the safety management level of industrial and mining enterprises, reduce accidents and injuries, and improve the working environment and quality of life of workers.

(3) The monitoring system's real-time monitoring and anomaly detection can detect potential environmental risks early, reduce the occurrence of industrial accidents, and play an important role in protecting the ecological environment and workers' health.

(4) In the long run, this research will promote the development of industrial and mining environmental monitoring technology, drive technological innovation and application in related fields, and promote economic development and social progress.

Based on the above positive effects, this study's over-reliance on advanced monitoring technology may lead to over-reliance on technology, thereby ignoring the importance of manual supervision and management. Good craftsmanship requires ensuring that technology is combined with human intervention to achieve the best results. The application of technology will lead to the reduction of some traditional positions, and training and job transfer measures need to be taken to reduce the negative impact on workers.

VI. CONCLUSION

This article uses multimodal deep learning and CLIP models to perform anomaly detection on the comprehensive environmental monitoring system of industrial and mining enterprises. Through experiments, an innovative solution combining multimodal LSTM models and CLIP models is proposed. Specifically, ResNet in the CLIP model can be used for image feature extraction, Transformer can be used for text feature encoding, and a multimodal LSTM model can be used for anomaly detection in the environment. In this study, the multimodal LSTM-CLIP model fully utilized multimodal information in environmental monitoring, and showed better accuracy in anomaly detection compared to the single-mode LSTM model, demonstrating good performance. Although this article has achieved some achievements, there are still some shortcomings. The complexity of this model and the high demand for computing resources are limited in resource constrained environments, and the experimental dataset is only conducted in coal mining enterprises, which is not rich enough. Future research can optimize the model structure, improve computational efficiency, and explore more types of multimodal data fusion methods to further enhance the performance and application scope of environmental monitoring systems. In addition, the model can be applied to the actual environment of other industrial and mining enterprises to verify its universality and stability.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CONFLICTS OF INTEREST

The authors confirm that they have no financial or non-financial competing interests with any companies, organizations, or individuals while preparing and submitting the manuscript.

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